Class Project (1 to 4 students)

Problem:
Implement a system that performs the operations in the table:

Operation code		Operation (Y)	Description
s1	s0		
0	0	A * B	Multiply A by B using parallel CSA and CPA
0	1	A * B	Multiply A by B using shift and add.
1	0	A - B	Simple subtraction
1	1	5 * (A+B)	Multiply the addition result by 5

Assuming all nMOS has (W= 4λ, L=2λ) and all pMOS has (W= 8λ, L=2λ).

The inputs and outputs are:

Inputs:
- An 8-bit input A (a7: a0)
- An 8-bit input B (b7: b0)
- A 2-bit selector S (s1: s0)
- A common VDD and a common GND lines
- Any other necessary inputs (specify them)

Outputs:
- A single 16-bit output Y (y15:y0)
- Possible carry-out bit
- Any other necessary outputs (specify them)

Requirement:

- Part 1: Verify the design using a logic design tool.
- Part 2: Implement your design using Magic VLSI layout tool to generate your project layout http://opencircuitdesign.com/magic/
- Part 3: Test your design using *irsim* to simulate your project. http://opencircuitdesign.com/irsim/

Deliverables:

1- One PDF that contains the following sections:
- Front page with group names and IDs.
- Introduction
- Problem statement and specifications
- Motivation
- Solution design using a logic design tool that you studied in the previous logic design courses
- Stick diagram for each component of your design. The building block components are enough.
- Testing strategy and results: they should show instructions to simulate and verify your design, by including Linux terminal commands for Magic and Irsim used to run your project with a few different inputs.

2- A compressed folder containing:
- Source code and layout.